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SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 1.01

GAT Mental Arithmetic, Divisibility & Surds

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

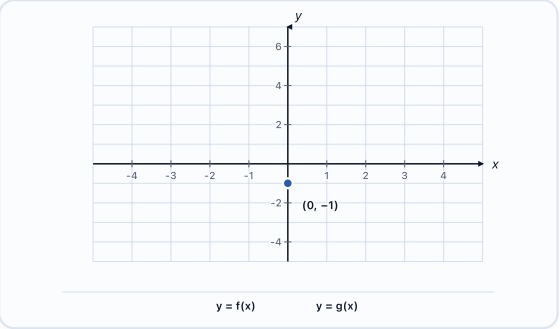
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 C	Q4 D	Q5 A
Q6 B	Q7 C	Q8 D	Q9 C	Q10 A
Q11 C	Q12 A	Q13 B	Q14 D	Q15 C
Q16 B	Q17 D	Q18 C	Q19 D	Q20 C
Q21 A	Q22 B	Q23 A	Q24 A	Q25 A

GAT Mental Arithmetic, Divisibility & Surds Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 6x + 20$ in the domain of GAT Arithmetic Shortcuts, Surds & Number Sense. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(5)$.



A 26 ✓

B 29

C 15

D 24

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Complete the square: $x^2 - 6x + 20 = (x-3)^2 - 9 + 20 = (x-3)^2 + 11$, giving minimum value $k = 11$. **[M1]**
2. Evaluate at $x = 5$: $f_1(5) = (2)^2 + 11 = 15$. **[M1]**
3. Sum the two quantities: $k + f_1(5) = 11 + 15 = 26$. **[A1]**

- **Final Verified Answer:** 26.

2 MCQ

A mathematical model in GAT Arithmetic Shortcuts, Surds & Number Sense is defined by the cubic function $y_2(x) = 2x^3 - 3x^2 + 15x - 8$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69

B 51 ✓

C 36

D 27

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 6x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 6(9) - 6(3) + 15 = 51$. **[M1][A1]**

- **Final Verified Answer:** 51.

3 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, two displacement vectors are given by $\mathbf{u}_3 = \langle 3, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A -29

B 73

C 44 ✓

D 156

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (3)(9) + (-8)(7) = 27 + (-56) = -29$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 3^2 + (-8)^2 = 73$. **[M1]**
3. Combine: $-29 + 73 = 44$. **[A1]**

- **Final Verified Answer:** 44.

4 MCQ

The linear growth profile shown in the diagram for GAT Arithmetic Shortcuts, Surds & Number Sense follows an arithmetic progression with first term $a_1 = 4$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 88

B 780

C 1320

D 690 ✓

WORKED SOLUTION & VERIFICATION

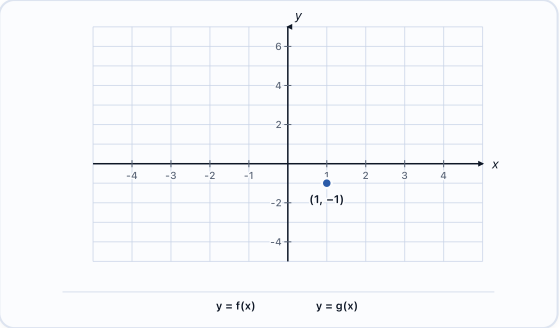
Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 4, d = 6$: $S_{15} = \frac{15}{2}[8 + 14 \times 6] = \frac{15}{2} \times 92 = 690$. **[M1][A1]**

- **Final Verified Answer:** 690.

5 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 12x)\,dx$.



A 72 ✓

B 96

C 84

D 37

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Integrate term by term: $\int (18x^2 + 12x)\,dx = [6x^3 + 6x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 6(2)^2) - 0 = 48 + 24 = 72$. **[M1][A1]**

- **Final Verified Answer:** 72.

6 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, consider the quadratic function $f_6(x) = x^2 - 8x + 99$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(11)$.

A 219

B 215 ✓

C 132

D 208

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Complete the square: $x^2 - 8x + 99 = (x-4)^2 - 16 + 99 = (x-4)^2 + 83$, giving minimum value $k = 83$. **[M1]**
2. Evaluate at $x = 11$: $f_6(11) = (7)^2 + 83 = 132$. **[M1]**
3. Sum the two quantities: $k + f_6(11) = 83 + 132 = 215$. **[A1]**

- **Final Verified Answer:** 215.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 4x^2 + 33x - 15$ studied in GAT Arithmetic Shortcuts, Surds & Number Sense. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 417	B 320
C 353 ✓	D 73

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 8x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 6(64) - 8(8) + 33 = 353$. **[M1][A1]**

- **Final Verified Answer:** 353.

8 MCQ

In GAT Arithmetic Shortcuts, Surds & Number Sense, two displacement vectors are given by $\mathbf{u}_8 = \langle 4, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A -3	B 65
C 132	D 62 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Arithmetic Shortcuts, Surds & Number Sense):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (4)(8) + (-7)(5) = 32 + (-35) = -3$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 4^2 + (-7)^2 = 65$. **[M1]**
3. Combine: $-3 + 65 = 62$. **[A1]**

- **Final Verified Answer:** 62.

9 MCQ

What is the units digit of 2^{17}?

A 6	B 8
C 2 ✓	D 4

WORKED SOLUTION & VERIFICATION

Powers of 2 repeat their units digits in a cycle of length 4: $(2, 4, 8, 6)$. Dividing the exponent 17 by 4 leaves remainder 1, giving units digit 2.

10 MCQ

What is the units digit of 3^{21}?

A 3 ✓	B 9
C 1	D 7

WORKED SOLUTION & VERIFICATION

Powers of 3 repeat their units digits in a cycle of length 4: $(3, 9, 7, 1)$. Dividing the exponent 21 by 4 leaves remainder 1, giving units digit 3.

11 MCQ

What is the units digit of 7^{23}?

A 9
B 7
C 3 ✓
D 1

WORKED SOLUTION & VERIFICATION

Powers of 7 repeat their units digits in a cycle of length 4: $(7, 9, 3, 1)$. Dividing the exponent 23 by 4 leaves remainder 3, giving units digit 3.

12 MCQ

What is the units digit of 8^{27}?

A 2 ✓
B 4
C 8
D 6

WORKED SOLUTION & VERIFICATION

Powers of 8 repeat their units digits in a cycle of length 4: $(8, 4, 2, 6)$. Dividing the exponent 27 by 4 leaves remainder 3, giving units digit 2.

13 MCQ

What is the units digit of 2^{29}?

A 8
B 2 ✓
C 4
D 6

WORKED SOLUTION & VERIFICATION

Powers of 2 repeat their units digits in a cycle of length 4: $(2, 4, 8, 6)$. Dividing the exponent 29 by 4 leaves remainder 1, giving units digit 2.

14 MCQ

What is the units digit of 3^{33}?

A 1
B 7
C 9
D 3 ✓

WORKED SOLUTION & VERIFICATION

Powers of 3 repeat their units digits in a cycle of length 4: $(3, 9, 7, 1)$. Dividing the exponent 33 by 4 leaves remainder 1, giving units digit 3.

15 MCQ

What is the units digit of 7^{35}?

A 1

B 7

C 3 ✓

D 9

WORKED SOLUTION & VERIFICATION

Powers of 7 repeat their units digits in a cycle of length 4: $(7, 9, 3, 1)$. Dividing the exponent 35 by 4 leaves remainder 3, giving units digit 3.

16 MCQ

What is the units digit of 8^{39}?

A 6

B 2 ✓

C 4

D 8

WORKED SOLUTION & VERIFICATION

Powers of 8 repeat their units digits in a cycle of length 4: $(8, 4, 2, 6)$. Dividing the exponent 39 by 4 leaves remainder 3, giving units digit 2.

17 MCQ

What is the units digit of 2^{41}?

A 6

B 4

C 8

D 2 ✓

WORKED SOLUTION & VERIFICATION

Powers of 2 repeat their units digits in a cycle of length 4: $(2, 4, 8, 6)$. Dividing the exponent 41 by 4 leaves remainder 1, giving units digit 2.

18 MCQ

What is the units digit of 3^{45}?

A 1

B 7

C 3 ✓

D 9

WORKED SOLUTION & VERIFICATION

Powers of 3 repeat their units digits in a cycle of length 4: $(3, 9, 7, 1)$. Dividing the exponent 45 by 4 leaves remainder 1, giving units digit 3.

19 MCQ

What is the units digit of 7^{47}?

A 9
B 7
C 1
D 3 ✓

WORKED SOLUTION & VERIFICATION

Powers of 7 repeat their units digits in a cycle of length 4: $(7, 9, 3, 1)$. Dividing the exponent 47 by 4 leaves remainder 3, giving units digit 3.

20 MCQ

What is the units digit of 8^{51}?

A 6
B 4
C 2 ✓
D 8

WORKED SOLUTION & VERIFICATION

Powers of 8 repeat their units digits in a cycle of length 4: $(8, 4, 2, 6)$. Dividing the exponent 51 by 4 leaves remainder 3, giving units digit 2.

21 MCQ

What is the units digit of 2^{53}?

A 2 ✓
B 6
C 4
D 8

WORKED SOLUTION & VERIFICATION

Powers of 2 repeat their units digits in a cycle of length 4: $(2, 4, 8, 6)$. Dividing the exponent 53 by 4 leaves remainder 1, giving units digit 2.

22 MCQ

What is the units digit of 3^{57}?

A 9
B 3 ✓
C 1
D 7

WORKED SOLUTION & VERIFICATION

Powers of 3 repeat their units digits in a cycle of length 4: $(3, 9, 7, 1)$. Dividing the exponent 57 by 4 leaves remainder 1, giving units digit 3.

23 MCQ

What is the units digit of 7^{59}?

- **A** 3 ✓
- **B** 7
- **C** 1
- **D** 9

WORKED SOLUTION & VERIFICATION

Powers of 7 repeat their units digits in a cycle of length 4: $(7, 9, 3, 1)$. Dividing the exponent 59 by 4 leaves remainder 3, giving units digit 3.

24 MCQ

What is the units digit of 8^{63}?

- **A** 2 ✓
- **B** 8
- **C** 6
- **D** 4

WORKED SOLUTION & VERIFICATION

Powers of 8 repeat their units digits in a cycle of length 4: $(8, 4, 2, 6)$. Dividing the exponent 63 by 4 leaves remainder 3, giving units digit 2.

25 MCQ

What is the units digit of 2^{65}?

- **A** 2 ✓
- **B** 6
- **C** 4
- **D** 8

WORKED SOLUTION & VERIFICATION

Powers of 2 repeat their units digits in a cycle of length 4: $(2, 4, 8, 6)$. Dividing the exponent 65 by 4 leaves remainder 1, giving units digit 2.